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REPLY: We thank Goh *et al* for their interest in our work and their suggestions regarding the potential impact of our model's performance in assessing the effects of peripapillary lesions, especially in populations where myopia is more prevalent.[1] While we agree that exploring how the model performs in conditions such as peripapillary atrophy or colobomas would expand its clinical utility, our study was limited by the data available for training and validation. We used images from the UK Biobank, which unfortunately does not include detailed annotations for peripapillary lesions.[2] Furthermore, at the time of our study, publicly accessible datasets that included labeled cup-to-disc ratio (CDR) associated with peripapillary lesions were unavailable. Nevertheless, if such data were accessible, direct validation or additional model fine-tuning would be invaluable.

To investigate the utility of a dynamically weighted (DW) loss function while including total gradable images (68 695) to enhance our regression model's performance and generalizability for quantifying CDR from fundus images, we first incorporated DW loss functions rather than dropping poorly predicted images (7780) during training. Each training sample's weight was assigned based on its prediction error. The base weight, determined by the inverse frequency of its corresponding CDR value, was increased by an additional term: alpha (α) multiplied by the error. We compared several loss functions, including standard mean squared error (MSE), mean absolute error, Huber loss, and DW MSELossFlat with different α values (0.5, 1.5, and 3.0).[3,4] Our experiments demonstrated that the DW MSELossFlat with $\alpha = 0.5$ achieved the lowest validation loss (9e-4), effectively addressing class imbalance by emphasizing challenging fundus images. Second, we systematically evaluated multiple optimizers (Adam, Ranger, and Stochastic Gradient Descent) and batch sizes (16, 32, 64, and 128). The learning rate finder was employed for each configuration to determine an optimal learning rate, and the model was fine-tuned over 3 epochs. The hyperparameters (DW MSELossFlat with $\alpha = 0.5$, Ranger optimizer, and batch size 128) that achieved the lowest validation loss were chosen for the final training phase. This final step in the training was the same as the original model used, as described in the Methods section of our paper,[5] except for employing the optimal optimizer, batch size, and a DW MSELossFlat with $\alpha = 0.5$ with a total of 68 695 fundus images. In our original model, we implemented a batch size of 64 with the Adam optimizer on the final data of 60 917 fundus images. A comparison was made between the weighted and original models with and without excluding poor-performing images. The weighted model, excluding the underperforming images, also accurately quantifies the CDR even in low-quality fundus images, overcoming challenges such as overexposed and underexposed optic discs and the presence of parapapillary alpha or beta zones. However, we limited the number of such images in the training set to fully learn features from the peripapillary or retinal lesions, which affected the accurate estimation of the CDR.

Applying the weighted loss function with optimal hyperparameters has not dramatically improved the coefficient of determination (R^2) compared to the original model, with and without excluding the poor-performing images. Using a y-range of 0 to 10, the original model achieved an R^2 value of 0.7141. In contrast, when we narrowed the y-range to 0 to 1, the model improved its R^2 to 0.7706, based on the CDR training data from all grading images. However, overall metrics suggested that the original model performed slightly better when excluding the underperforming images. We retrained the weighted model using excluded data, resulting in an R^2 of 0.8496. The mean absolute error and MSE were 0.0552 (0.0544, 0.0560) and 0.0048 (0.0047, 0.0050), respectively, providing 95% confidence intervals. When the data are highly imbalanced (e.g., in our training dataset, the ratio of class 3−class 9 is approximately 185:1), the weight loss function becomes suboptimal. Thus, we dropped the underperforming images to train our original model, mainly those of fundus images with artifacts (which had a higher loss).

In conclusion, selecting the hyperparameters to train a model using the optimal batch size, optimizer, and weighted-loss function, especially when the data are imbalanced, could enhance the model's robustness and generalizability. Our model accurately estimates the CDR in low-quality fundus images, effectively addressing cases with peripapillary atrophy. However, validating our model on images with established peripapillary lesions would be useful before using it clinically. Future studies should aim to incorporate a wide range of data from multiethnic populations with disc abnormalities to further develop precise artificial intelligence-based models for quantifying the CDR.

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